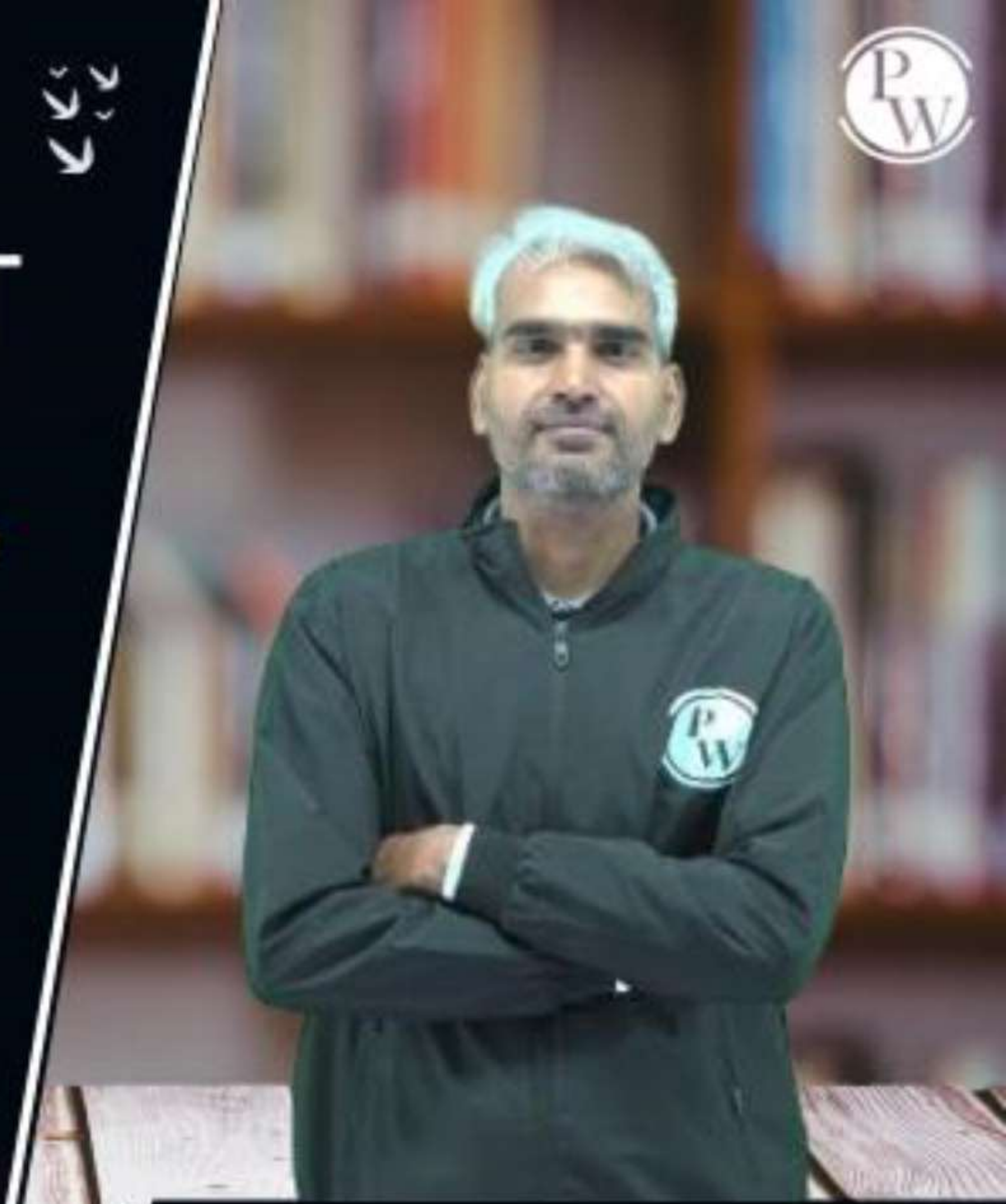
Computer Science & IT

Discrete Mathematics

Set Theory & Algebra

Lecture No. 09



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Recap of Previous Lecture

Topic Equivalence class

Topic Partition of a set

Topic Number of equivalence relation

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Topics to be Covered

Topic — Bell number

Topic — Number of equivalence relations

Topic — Partial Order Relation and Partially Ordered Set

Topic — Totally Ordered Set and Total Order Relation

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H.W. 6 Let $A = \{a, b, c, d, e, f\}$ i.e. $|A| = 6$

How many partitions are there of set A ?

$$\boxed{\text{if, } |A| = 6 \text{ then number of partitions of set } A = ?}$$

(1+5)

$|A| = 6 \Rightarrow$

$1+1+1+1+1+1 = \frac{{}^6C_1 * {}^5C_1 * {}^4C_1 * {}^3C_1 * {}^2C_1 * {}^1C_1}{6!} = 1$

$1+1+1+1+2 = \frac{{}^6C_1 * {}^5C_1 * {}^4C_1 * {}^3C_1}{4!} * {}^2C_2 = 15$

$1+1+1+3 = \frac{{}^6C_1 * {}^5C_1 * {}^4C_1}{3!} * {}^3C_3 = 20$

$1+1+2+2 = \frac{{}^6C_1 * {}^5C_1}{2!} * \frac{{}^4C_2 * {}^2C_2}{2!} = 45$

$1+1+4 = \frac{{}^6C_1 * {}^5C_1}{2!} * {}^4C_4 = 15$

$1+2+3 = {}^6C_1 * {}^5C_2 * {}^3C_3 = 60$

$1+5 = {}^6C_1 * {}^5C_5 = 6$

$2+2+2 = \frac{{}^6C_2 * {}^4C_2 * {}^2C_2}{3!} = 15$

$2+4 = {}^6C_2 * {}^4C_4 = 15$

$3+3 = \frac{{}^6C_3 * {}^3C_3}{2!} = 10$

$6 = {}^6C_6 = 1$

∴ Total no. of Partitions $= 1 + 15 + 20 + 45 + 15 + 60 + 6 + 15 + 15 + 10 + 1$

Ans 203

H.W. 6 Let $A = \{a, b, c, d, e, f\}$ i.e. $|A| = 6$

How many partitions are there of set A ?

$$\boxed{\text{if, } |A| = 6 \text{ then number of partitions of set } A = 203}$$

If $|A| = 6$, then no. of Equivalence relation on set $A = \boxed{203}$

Topic : Bell Number

Bell number B_n will represent the number of partions of a set A of size = n.

↳ i.e., B_n will represent the number of Equivalence relation on a set A of cardinality = n

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$B_0 = 1$

$B_1 = 1$

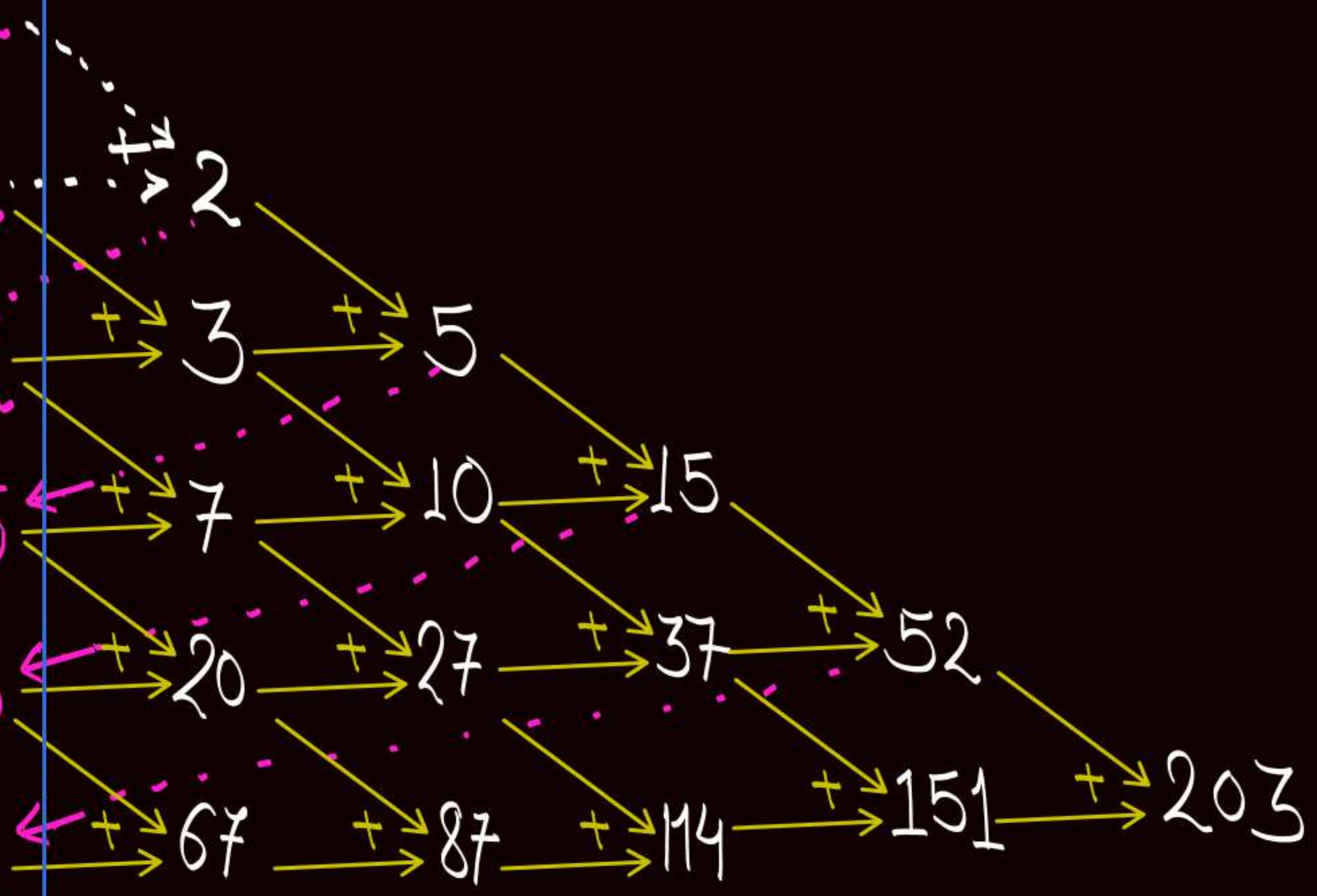
$B_2 = 2$

$B_3 = 5$

$B_4 = 15$

$B_5 = 52$

$B_6 = 203$



Q:→ Let $A = \{a, b, c, d, e\}$ i.e. $|A| = 5$.

How many Equivalence relation are there on set A such that the cardinality of each equivalence relation is exactly "9". { i.e. Number of order pairs in each such equivalence relation must be "9" }

$(1+4)$

$|A| = 5 \Rightarrow$

$1+1+1+1+1$

$(1\times1)+(1\times1)+(1\times1)+(1\times1)+(1\times1) = 5$ order pairs

$1+1+1+2$

$(1\times1)+(1\times1)+(1\times1)+(2\times2) = 7$ order pair

$1+1+3$

$(1\times1)+(1\times1)+(3\times3) = 11$ order pairs

$1+2+2$

$(1\times1)+(2\times2)+(2\times2) = 9$ order pairs $\Rightarrow \dfrac{{}^5C_1 * {}^4C_2 * {}^2C_2}{2!} = 15$

"15" such equivalence relation are there, with Exactly '9' order pairs

$1+4$

$(1\times1)+(4\times4) = 17$ order pairs

$2+3$

$(2\times2)+(3\times3) = 13$ order pairs

5

(5×5) —— 25 order pairs

Q: if $|A| = 6$, then How many Equivalence relation are there on set A, such that Cardinality of each equivalence relation is exactly '10'.

$(1+5)$

$|A| = 6 \Rightarrow$

No. of order pair

6 : $1+1+1+1+1+1 = \dfrac{{}^6C_1 * {}^5C_1 * {}^4C_1 * {}^3C_1 * {}^2C_1 * {}^1C_1}{6!} = 1$

8 : $1+1+1+1+2 = \dfrac{{}^6C_1 * {}^5C_1 * {}^4C_1 * {}^3C_1}{4!} * {}^2C_2 = 15$

12 : $1+1+1+3 = \dfrac{{}^6C_1 * {}^5C_1 * {}^4C_1}{3!} * {}^3C_3 = 20$

10 : $1+1+2+2 = \dfrac{{}^6C_1 * {}^5C_1}{2!} * \dfrac{{}^4C_2 * {}^2C_2}{2!} = 45$

18 : $1+1+4 = \dfrac{{}^6C_1 * {}^5C_1}{2!} * {}^4C_4 = 15$

14 : $1+2+3 = {}^6C_1 * {}^5C_2 * {}^3C_3 = 60$

26 : $1+5 = {}^6C_1 * {}^5C_5 = 6$

12 : $2+2+2 = \dfrac{{}^6C_2 * {}^4C_2 * {}^2C_2}{3!} = 15$

20 : $2+4 = {}^6C_2 * {}^4C_4 = 15$

18 : $3+3 = \dfrac{{}^6C_3 * {}^3C_3}{2!} = 10$

36 : $6 = {}^6C_6 = 1$

"45" Equivalence relation on set A. Such that the Cardinality of each equivalence relation is exactly $= 10$

Partial order relation

&

POSET

Topic : Partial Order Relation

- A relation R on set A is called a Partial order relation if and only if relation R is Reflexive, anti-symmetric <u>and</u> transitive

$$\boxed{\text{Partial order} = \text{Reflexive} + \text{Anti-symmetric} + \text{Transitive}}$$

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eg: Let $A = \{1, 2, 3\}$

$\Delta_A = \{(1,1), (2,2), (3,3)\}$

Reflexive ✓
Anti-symmetric ✓
Transitive ✓

∴ Diagonal relation is always a Partial order relation

* Diagonal relation on a set A is the smallest Partial order relation on set A

* Diagonal relation on set A is the only relation which is an Equivalence relation as well as a Partial order relation

eg. Let $A = \{1, 2, 3\}$

transitive as well

$$R = \{ \underbrace{(1,1), (2,2), (3,3)}_{\text{Reflexive}}, \underbrace{(1,2), (1,3), (2,3)}_{\text{Anti-symmetric}} \}$$

"R" is a partial order relation on set A.

Topic : Partially Ordered Set

[POSET]

- Let R is a partial order relation on set A.
- Set A along with the partial order relation R defined on set A is called a POSET

Note: If it is given that (A, R) is a POSET, then 'A' is a set and 'R' is a partial order relation on set A.

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eg: Let 'A' is any set of real numbers, then $(A, \leq)$ is a POSET.

Check whether the statement is true or false? { True ✓ }

$A = \{1, 2.5, 3\}$

$\leq = \{(1,1), (1,2.5), (1,3), \cancel{(2.5,1)}, (2.5,2.5), (2.5,3), \cancel{(3,1)}, \cancel{(3,2.5)}, (3,3)\}$ ⇒ Reflexive ✓, Anti-symmetric ✓, Transitive ✓

let $a, b \in A$

$(a,b) \in R$ iff $a \, R \, b$

Let relation 'R' is "$\leq$"

$\therefore (a,b) \in \leq$ iff $a \leq b$

"$\leq$" is a partial order relation on any set 'A' of real numbers $\therefore (A, \leq)$ is a POSET

eg:- Let A is any set of positive integers, then check whether $(A, \div)$ is a POSET or not?
where "$\div$" is relation divides {not divided by}
{ i.e. $(a, b) \in \div$ iff a divides b }

* (i) Every positive integer 'a' divides itself. $\therefore (a, a) \in \div, \forall a \in A$

$\boxed{\text{Hence, Reflexive}}$

(ii) let $a, b \in A$ such $a \neq b$, then
either $a \div b$ or $b \div a$ or none divides none
but ($a \div b$ and $b \div a$) can not be true at the same time

Hence, Relation divides($\div$) is anti-symmetric

(iii) if $((a \div b)$ and $(b \div c))$, then $(a \div c)$ is always true

Hence, Relation divides($\div$) is transitive

* Relation "$\div$" is a partial order relation on any set of positive integers A, $\therefore (A, \div)$ is a POSET.

eg: Let "C" is any Collection of sets.

i.e, $C = \{ S_1, S_2, S_3, \ldots \}$ where $S_1, S_2, \ldots$ all are sets.

then $(C, \subseteq)$ is a POSET

Relation is subset

1. Every set is a subset of itself ∴ Reflexive
2. if $S_1, S_2 \in C$ and $S_1 \neq S_2$, then Both S_1 and S_2 can not be subset of each other at the same time, ∴ Anti-Symmetric
3. If $S_1 \subseteq S_2$ and $S_2 \subseteq S_3$ then $S_1 \subseteq S_3$ is always true ∴ Transitive.

Note :-

If A is any set of integers,
then $(A, \div)$ may or may not be a POSET,

because it may be the case that $0 \in A$
and '0' does not divide '0' $\therefore (0,0) \notin \div$

Hence not reflexive

Topic : Comparability

- Let (A, R) be a POSET,
- Let $a, b \in A$,

  Elements 'a' and 'b' are said to be comparable if and only if,

$$a \, R \, b \quad \underline{\text{or}} \quad b \, R \, a$$

$$\underline{\text{i.e.}}, \ (a,b) \in R \quad \underline{\text{or}} \quad (b,a) \in R$$

Let $a, b \in A$,
If neither $(a,b) \in R$ nor $(b,a) \in R$, then a & b are not Comparable w.r.t. POSET (A,R)

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* In a POSET (A, R),

R is a partial order relation,

i.e. R is guaranteed to be reflexive

$\therefore (a, a) \in R \;\forall\, a \in A$

Hence, Every element of set A is always Comparable with itself.

eg: Let $A = \{1, 2, 3, 4\}$
and $(A, \div)$ is POSET.

* '1' divides Every element of the set.
  ∴ '1' relates with Every element of the set
  Hence '1' is comparable with all the elements of the set

* 2 is comparable with 1, because $(1, 2) \in \div$
* 2 is comparable with 2, because $(2, 2) \in \div$
* 2 is comparable with 4, because $(2, 4) \in \div$

* "2 & 3" are not comparable because neither 2 divides 3 (i.e. $(2,3) \in \div$)
  nor 3 divides 2 (i.e. $(3,2) \in \div$)

Similarly, '3 & 4' are not comparable.
because neither $(3,4) \in \div$ nor $(4,3) \in \div$

* **Note :-**

In a POSET (A, R)

Every pair of elements of set A

need not be Comparable

Topic : Totally Ordered Set / (TOSET)

- Let (A, R) be a POSET,

  If Every pair of elements of set A are Comparable, then POSET (A, R) is called a TOSET and relation 'R' can be called a "Total order relation"

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Note :-

① Every TOSET is also a POSET, But Every POSET need not be a TOSET

Similarly, ② Every Total order relation is also a Partial order relation, but Every Partial Order relation need not be a Total order relation.

eg: Let $A_1 = \{1, 2, 3, 4\}$
and $(A_1, \div)$ is a POSET

- We know, that there are some pair of elements in set A_1 that are not comparable w.r.t. POSET $(A_1, \div)$

$\therefore$ $(A_1, \div)$ is just a POSET, but not a TOSET.

eg: Let $A_2 = \{1, 2, 4, 8\}$
and $(A_2, \div)$ is a POSET.

- Every pair of elements of set A_2 are comparable

$\therefore$ $(A_2, \div)$ is a TOSET

- $(A_2, \div)$ is a TOSET, but it is also a POSET.

2 mins Summary

- **Topic** Bell number
- **Topic** Number of equivalence relations
- **Topic** Partial Order Relation and Partially Ordered Set
- **Topic** Totally Ordered Set and Total Order Relation

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THANK - YOU

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